

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1003+060

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

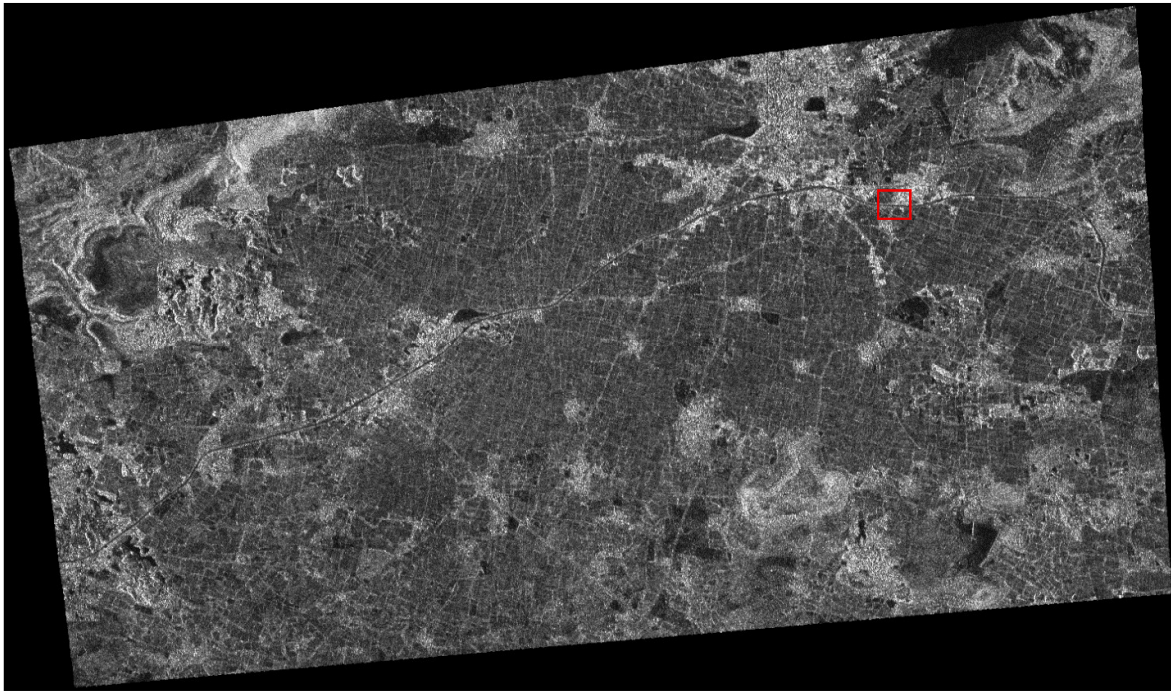
Flight Direction - DESCENDING

Polarization – VV+VH

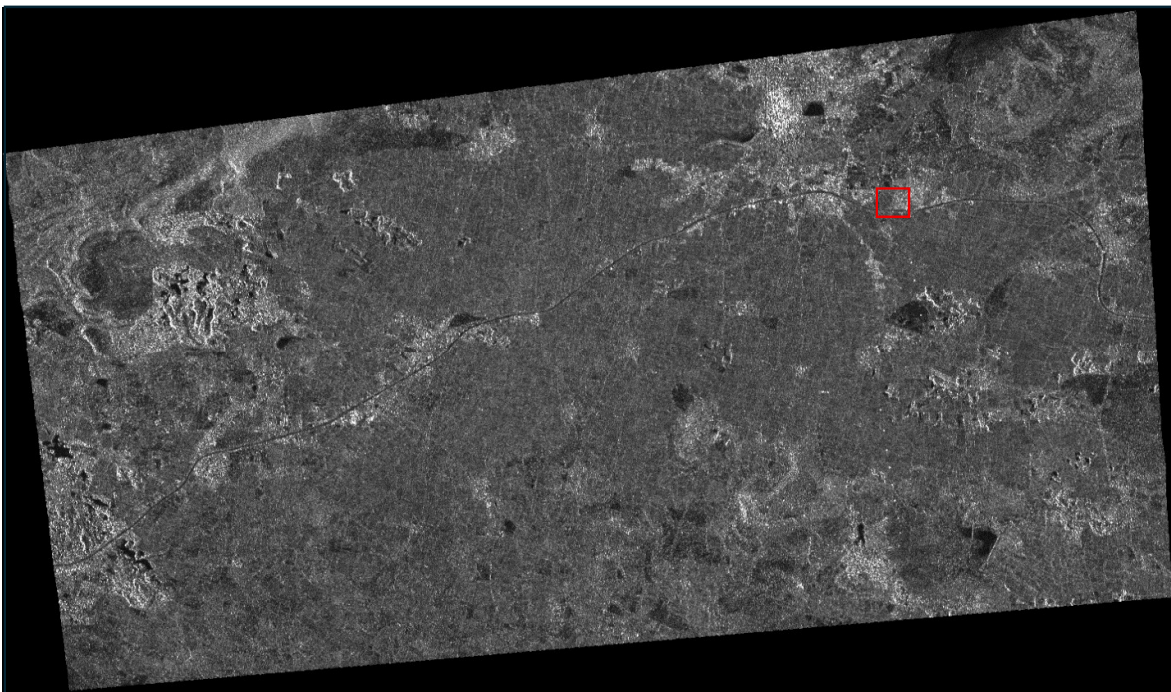
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

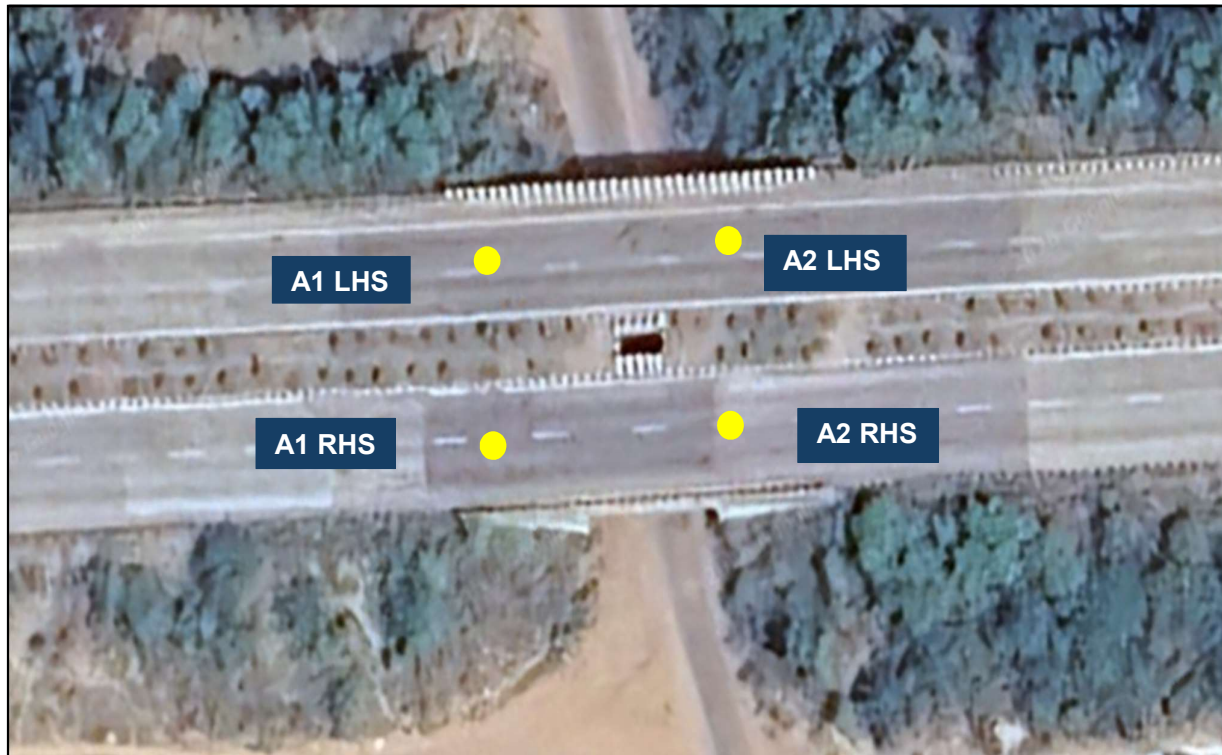


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

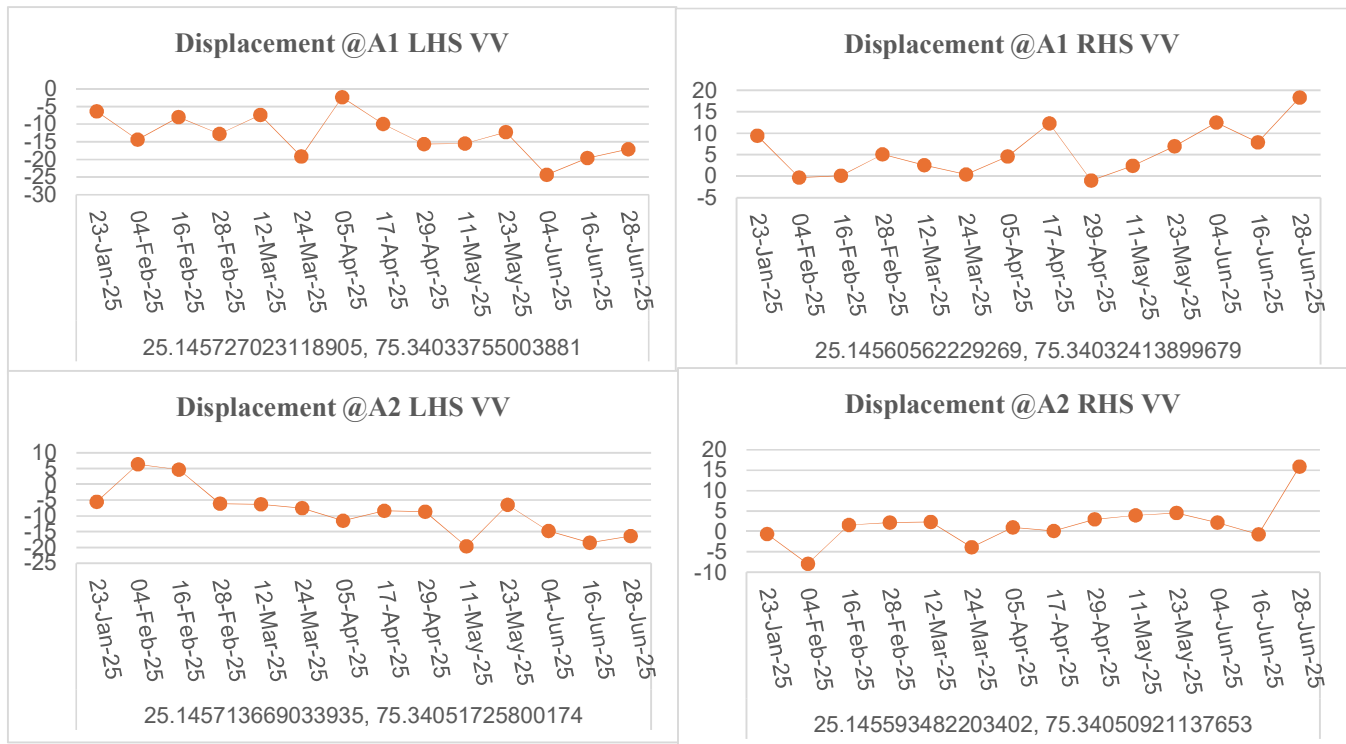
VUP – CH-1003+060

Length – 30.78M

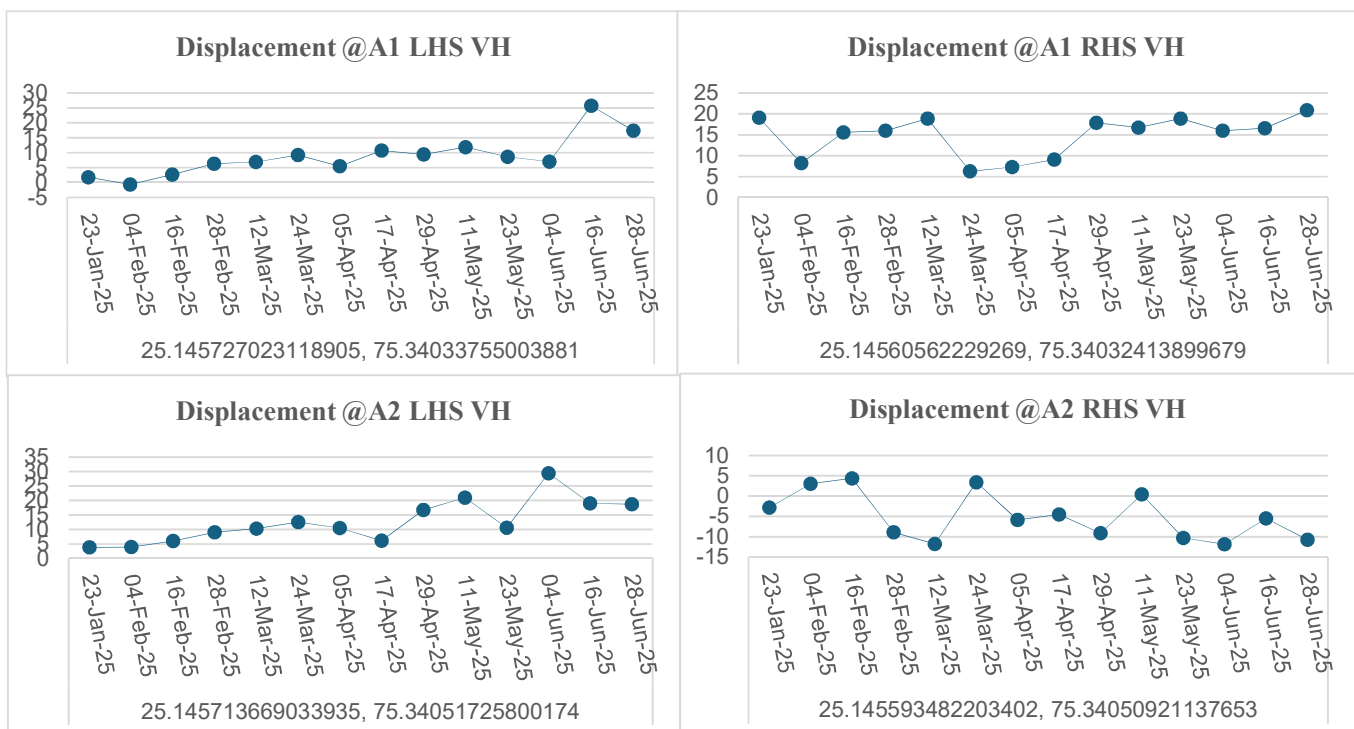


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



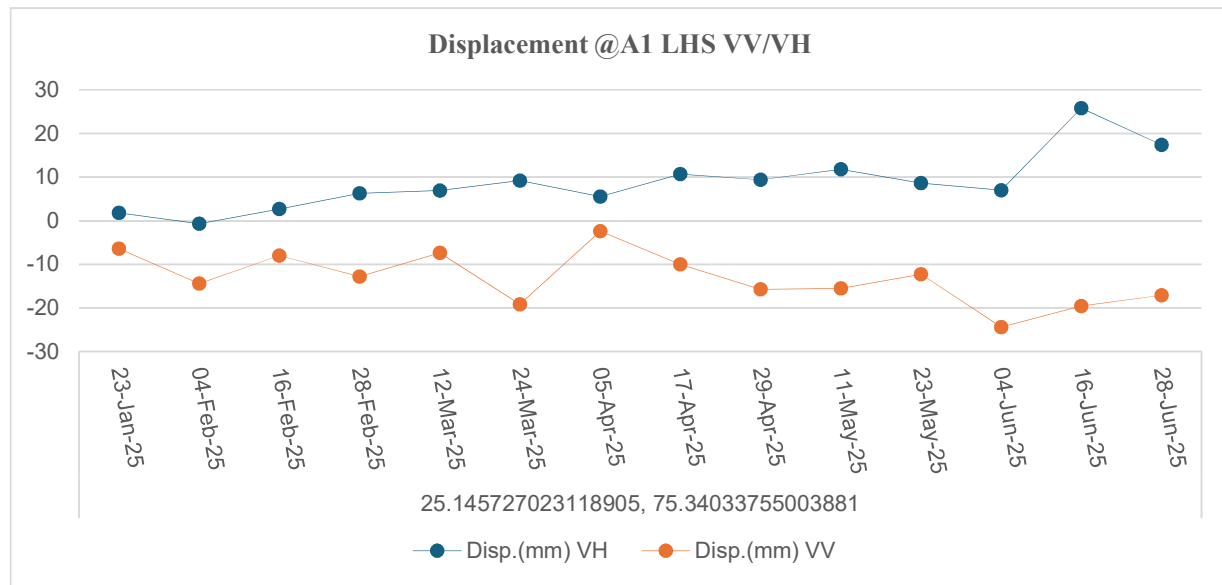
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.145727023118905, 75.34033755003881)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** is consistently negative, indicating continuous subsidence. Magnitudes fluctuate between -7 mm and -25 mm, with increasing subsidence intensity after April.
- **Horizontal Displacement (VH)** remains consistently positive throughout the monitoring period, showing a gradual upward trend from +1 mm (Jan) to a peak of +25 mm in mid-June 2025.

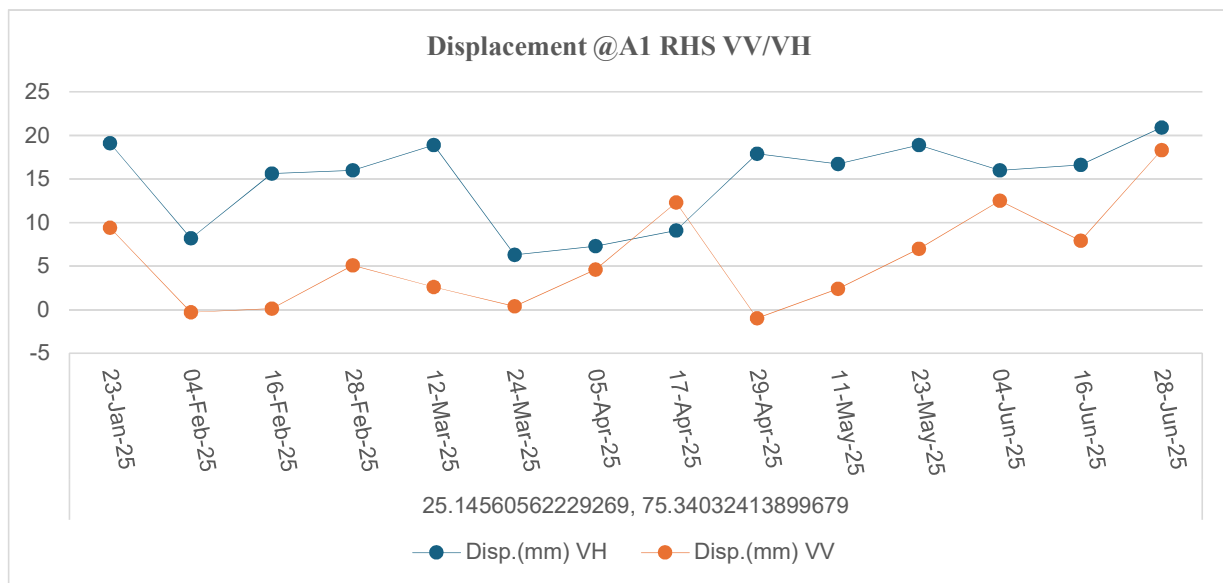
Critical Events & Anomalies:

- **February 2025:** A short-term increase in subsidence (-15 mm) coincides with only minor VH change, suggesting vertical-dominant deformation.
- **June 2025:** VH decreases slightly, but VV remains strongly negative, indicating subsidence persistence even as lateral motion stabilizes.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.14560562229269, 75.34032413899679)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values remain consistently positive throughout the monitoring period, ranging approximately from +6 mm to +21 mm.
- **Vertical displacement (VV)** values are predominantly positive, indicating net uplift. Magnitudes vary from 0 mm to +18 mm, with a clear increasing uplift trend after April 2025.

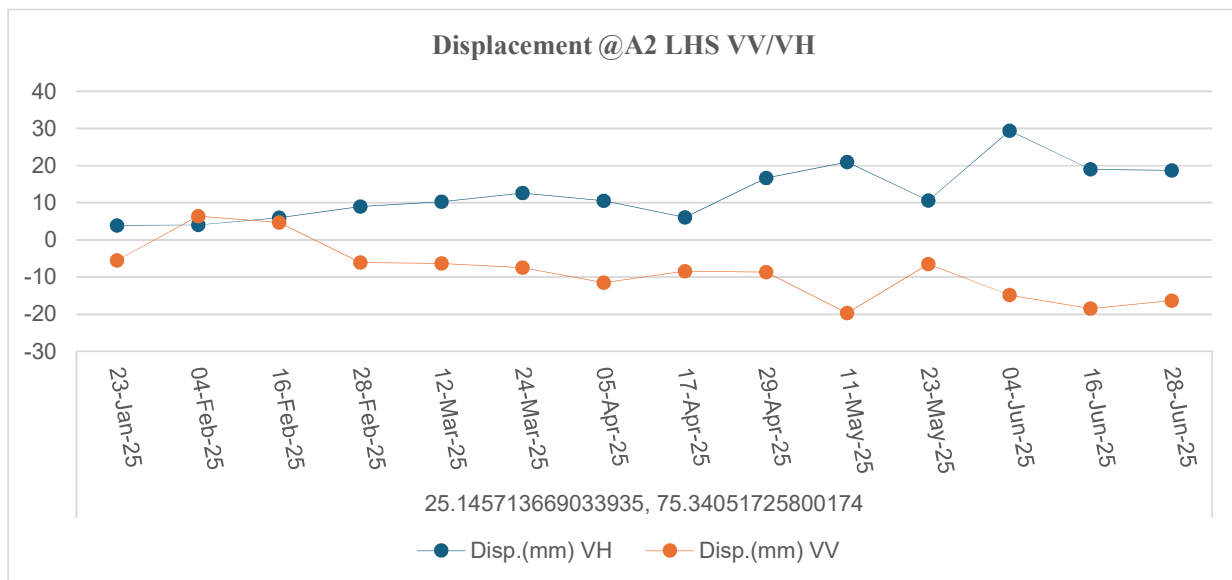
Critical Events & Anomalies:

- **February 2025:** A short-term drop in VH (+8 mm) occurs while VV remains near zero, suggesting temporary lateral relaxation without vertical response.
- **June 2025:** VH reaches its maximum (+21 mm) concurrently with a significant uplift peak (+18 mm) representing the strongest deformation phase observed at A1 RHS.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.145713669033935, 75.34051725800174)



Key Observations:

Displacement Magnitude Patterns:

- Horizontal displacement (VH)** remains consistently positive across the entire monitoring period, increasing from +3–5 mm (Jan 2025) to +18–30 mm.
- Vertical Displacement (VV)** values are predominantly negative, ranging approximately from –5 mm to –22 mm, indicating continuous vertical subsidence. No sustained uplift phases are observed.

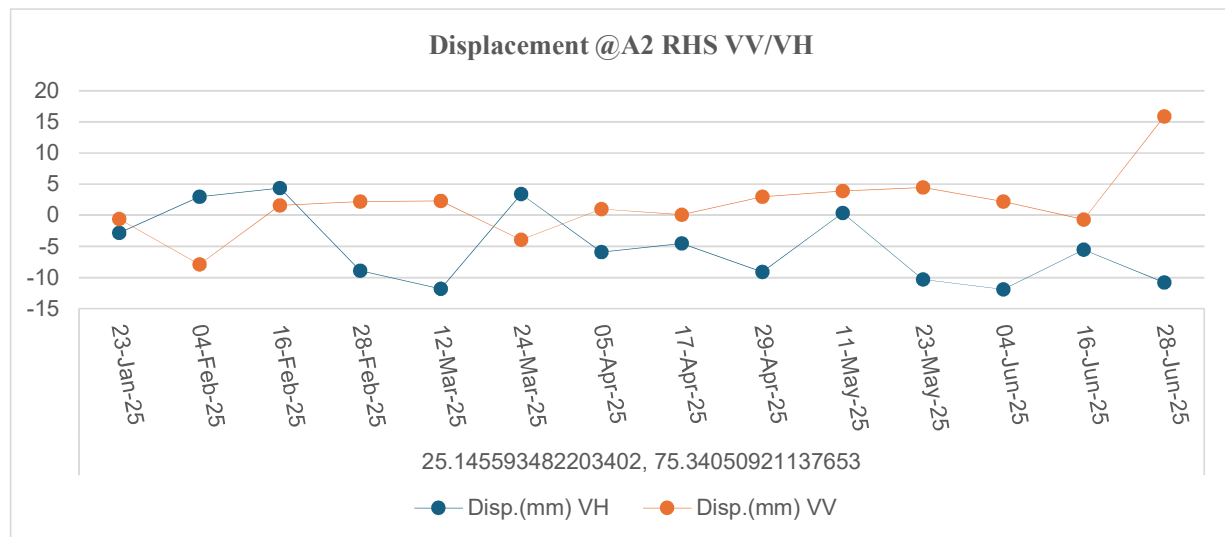
Critical Events & Anomalies:

- February 2025:** A brief vertical uplift (+5 mm) occurs while VH remains low and positive, suggesting a short-lived vertical rebound without lateral intensification.
- June 2025:** VH reaches its absolute maximum (+30 mm) concurrently with continued subsidence (–15 mm), signalling high structural strain.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.145593482203402, 75.34050921137653)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** displays a gradual upward tendency over time, with values varying from about -8 mm settlement to +16 mm uplift.
- **Horizontal (VH) displacement** at A2 RHS shows pronounced oscillatory behaviour with values ranging approximately from -12 mm to +4 mm.

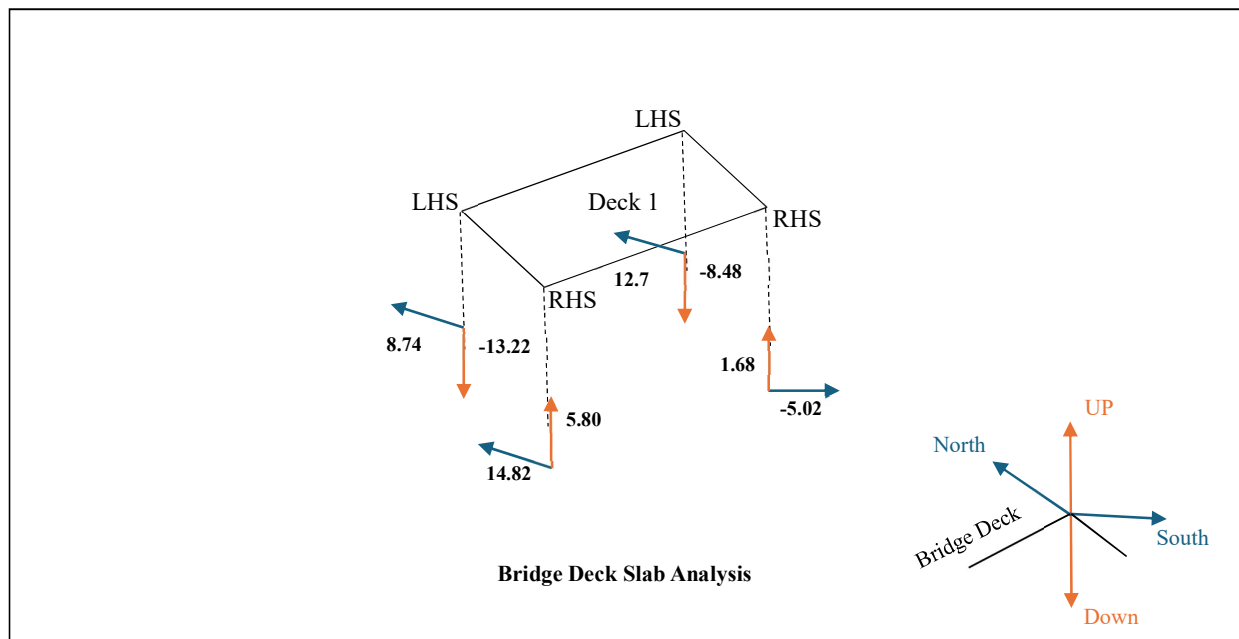
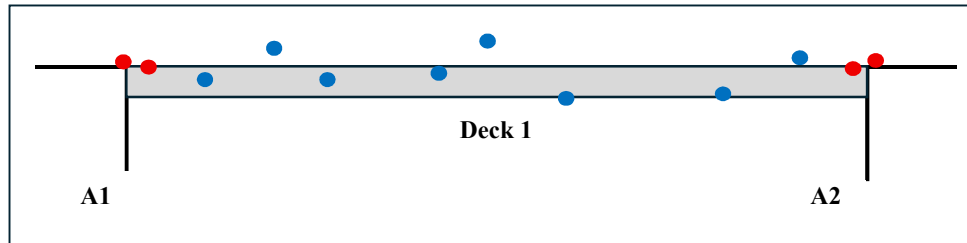
Critical Events & Anomalies:

- **January-February 2025:** VH starts slightly negative (-3 mm) and briefly rises to +4 mm, while VV fluctuates between -8 mm and +2 mm, indicating initial instability with mixed vertical response.
- **March 2025:** VH experiences a sharp drop to around -12 mm, while VV remains close to neutral highlighting a significant lateral displacement event without corresponding vertical amplification.
- **June 2025:** VV shows a pronounced peak reaching +16 mm uplift, while VH stays negative, indicating dominant vertical uplift accompanied by persistent lateral offset.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits clear non-uniform vertical displacement across the LHS and RHS supports. The central RHS deck location records a downward displacement of approximately -8.48 mm, suggesting sagging behaviour beneath the deck slab.

Horizontal Displacement Behaviour:

- Horizontal displacement magnitudes demonstrate strong directional variability across the deck slab. The LHS upper region records a horizontal displacement of approximately $+8.74$ mm, indicating lateral movement toward the indicated direction.

CONCLUSION:

- The bridge deck slab demonstrates a highly non-symmetric displacement response, governed by combined vertical bending and torsional effects. Continued monitoring is recommended to assess whether this deformation pattern represents a transient response or an evolving structural behaviour.